PE2

CS2030S AY23/24 Sem 2

plab3195

1 (2.0/2.0 marks)

```java
// @author A0288236U

// i can't believe this was it ;-;
public interface Valuable {
  public double getPrice();
}
```

Your submitted code passed all our test cases.

2 (5.0/5.0 marks)

```java
// @author A0288236U

public class Book {
  private String title;
  private String genre;
  private Valuable priceFunc;

  public Book(String title, String genre, Valuable priceFunc) {
    this.title = title;
    this.genre = genre;
    this.priceFunc = priceFunc;
  }

  public String getTitle() {
    return this.title;
  }

  public String getGenre() {
    return this.genre;
  }

  public double getPrice() {
    return this.priceFunc.getPrice();
  }

  public boolean hasTitle(String t) {
    return this.title.equals(t);
  }

  @Override
  public String toString() {
    return String.format("%s (S$ %.2f)", this.getTitle(), this.getPrice());
```

```
  }
}
```

Your submitted code passed all our test cases.

3 (8.0/8.0 marks)

```
// @author A0288236U
import java.util.ArrayList;
import java.util.List;

public class BookShop<T extends Book> {
  private List<? extends T> bookList;

  public BookShop(List<? extends T> bookList) {
    this.bookList = bookList;
  }

  public BookShop() {
    this.bookList = new ArrayList<>();
  }

  public List<? extends T> getBooks() {
    return this.bookList;
  }

  public List<? extends T> findBooksByGenre(String genre) {
    return this.bookList.stream().filter(x -> x.getGenre().equals(genre)).toList();
  }

  public boolean hasBook(String title) {
    return this.bookList.stream().filter(x -> x.hasTitle(title)).count() > 0;
  }

  @Override
  public String toString() {
    return String.format("%s\n",
        this.getBooks().stream()
        .reduce("", (x, y) -> String.format("%s\n- %s", x, y), (x, y) -> x + y));
  }
}
```

Your submitted code passed all our test cases.

4 (6.0/6.0 marks)

```
// @author A0288236U
import java.util.List;
import java.util.stream.Stream;

public class Q4 {
  public static <T extends Book, U extends Book>
      BookShop<Book> mergeShops(BookShop<T> b1, BookShop<U> b2) {
```

```
    return new BookShop<Book>(
        Stream.concat(b1.getBooks().stream(), b2.getBooks().stream()).toList());
  }
}
```

Your submitted code passed all our test cases.

5 (6.0/6.0 marks)

```
// @author A0288236U
import java.util.List;
import java.util.stream.Stream;

public class Q5 {
  public static List<String> findUniqueGenres(List<BookShop<? extends Book>> bookShopList) {
    return bookShopList.stream().map(x -> x.getBooks().stream())
      .reduce(Stream.empty(), (x, y) -> Stream.concat(x, y))
      .map(x -> x.getGenre()).distinct().sorted().toList();
  }
}
```

Your submitted code passed all our test cases.

6 (6.0/6.0 marks)

```
// @author A0288236U
import java.util.List;
import java.util.stream.Stream;

public class Q6 {
  public static List<BookShop<Book>>
      consolidateShopsbyGenre(List<BookShop<? extends Book>> bookShopList) {
    return Q5.findUniqueGenres(bookShopList).stream()
      .map(genre -> bookShopList.stream().map(bs -> bs.findBooksByGenre(genre).stream())
          .reduce(Stream.empty(), (x, y) -> Stream.concat(x, y)))
      .map(x -> new BookShop<Book>(x.toList())).toList();
  }
}
```

Your submitted code passed all our test cases.

7 (6.0/6.0 marks)

```
// @author A0288236U
import java.util.List;
import java.util.stream.Stream;

class Q7 {
  static class DiscountedBook extends Book {
    public DiscountedBook(Book b, double discount) {
      super(b.getTitle(), b.getGenre(), () -> b.getPrice() * (100 - discount) / 100);
    }
  }
```

```
  public static List<BookShop<DiscountedBook>>
      discountRemainingStock(List<BookShop<? extends Book>> bookShopList, long k) {
    return bookShopList.stream().filter(bs -> bs.getBooks().size() < k)
      .map(bs -> new BookShop<DiscountedBook>(
            bs.getBooks().stream().map(b -> new DiscountedBook(b, 50)).toList())).toList();
  }
}
```

Your submitted code passed all our test cases.

8 (6.0/6.0 marks)

```
// @author A0288236U
import java.util.Comparator;
import java.util.List;
import java.util.stream.Stream;

class Q8 {
  static class ShopComparator<T extends Book> implements Comparator<BookShop<? extends T>> {
    public int compare(BookShop<? extends T> stall1, BookShop<? extends T> stall2) {
      return (int) (stall1.getBooks().stream()
          .map(Book::getPrice)
          .reduce(Double.MAX_VALUE, Math::min)
          - stall2.getBooks().stream()
          .map(Book::getPrice)
          .reduce(Double.MAX_VALUE, Math::min));
    }
  }

  public static <T extends Book> List<BookShop<T>>
      searchForBook(List<BookShop<? extends T>> bookShopList, String title) {
    return bookShopList.stream().filter(bs -> bs.hasBook(title))
      .sorted(new ShopComparator<T>()).map(
          bs -> new BookShop<T>(
            bs.getBooks().stream().filter(b -> b.hasTitle(title)).toList())).toList();
  }
}
```

Your submitted code passed all our test cases.

Summary

Question	Max Marks	Marks
1	2.0	2.0
2	5.0	5.0
3	8.0	8.0
4	6.0	6.0
5	6.0	6.0
6	6.0	6.0
7	6.0	6.0

Question	Max Marks	Marks
8	6.0	6.0
Style		-0
TOTAL	45	**45.0**

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